



Norwegian University of
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Formally Verified Reduction Proof for the Free XOR Garbling Scheme

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Problem description

Garbled circuits are a fundamental cryptographic primitive used in secure two-party computation (2PC), enabling two parties to jointly compute a function without revealing their private inputs—only the output is revealed. The concept was introduced by Andrew Yao in 1986 [Yao86], where functions are expressed as Boolean circuits and each wire is labelled with encrypted wire keys. Early constructions required four ciphertexts per gate, resulting in significant communication overhead and limiting practical adoption.

Subsequent optimizations such as Point-and-Permute, Half-Gates and the Free XOR technique greatly reduced the number of ciphertexts needed per gate, as well as reducing the number of gates to be transmitted, leading to substantial performance improvements [BMR90; ZRE15; KS08]. These optimizations form the basis of many state-of-the-art secure computation frameworks.

Formal verification of cryptographic constructions is essential to ensure correctness and prevent subtle vulnerabilities arising from unchecked corner cases. Brzuska and Oechsner [BO23] developed a state-separating pen-and-paper reduction proof for Yao’s original garbling scheme. State-separating proofs model protocol components as isolated “packages” with private state and oracle interfaces, allowing for structured security reductions. Their work established security by reducing the garbling scheme to the IND-CPA security of the encryption primitive used for gate tables.

Domino is an automated verification tool specifically designed for state-separating proofs. It has been used to reason about complex real-world protocols such as TLS [Raj25] and WPA2 [BDE+22]. Domino offers a high level of automation and readability because proof scripts resemble the pseudocode found in traditional cryptographic proofs.

Furthermore, the above mentioned proof for Yao’s garbling scheme by Brzuska and Oechsner [BO23] was later formalized in Domino [BW]. It can serve as a good starting point for the thesis to observe how games may be converted from pen and paper proofs to Domino code.

The goal of this thesis is to formalize and mechanize a security reduction for the Half-Gates garbling scheme within the Domino framework. This includes:

- Developing a state-separating, pen-and-paper reduction proof for the Free XOR technique.
- Encoding this proof in Domino.

This work extends formal verification techniques to an optimized and widely deployed garbling construction and demonstrates Domino’s applicability to modern

cryptographic primitives.

Approved: 2026-03-09 – Tjerand Silde, NTNU (Main supervisor)

Abstract

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This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

After this fourth paragraph, we start a new paragraph sequence. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of

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Preface

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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TODO: Remember to change minus to hyphen where needed.

Chapter 1

Introduction

Secure Multiparty Computation, or just Multi-Party Computation (MPC) refers to a class of problems in which two or more parties want to compute a common function using their individual inputs, but do not want to share these inputs with each other. Common use cases of MPC today include running heavy computation on cloud servers, or querying third party datasets that cannot be shared directly.

Since remote servers are often owned and operated by a separate entity, it is difficult to obtain the same security and privacy guarantees that a user's own machine can provide. Besides user requirements, cybersecurity regulations also require methods to secure the data being processed from the data processor, such as in the case of private healthcare data, or other forms of personal data [16].

Two-party computation (2PC) is a special case of MPC in which only two parties are involved in the protocol. Often attributed to Andrew Yao's seminal work in 1986 [Yao86], the first approach for 2PC was to represent the function to be computed as a boolean circuit, and to use a Garbled Circuit (GC) to compute the output of the function, while maintaining the secrecy of the inputs of both parties. For more than a decade after this work, Garbled Circuits were considered to not be practical because of the high bandwidth and computation costs of garbling, and due to the fact that garbling an arbitrary function into an optimized boolean circuit is difficult. However, there has been ongoing work in the field that has led to optimizations that allow them to be somewhat usable in generic 2PC systems like Fairplay [MNPS04], and in niche applications such as smart contracts [HYLL].

In this work, we aim to convert the proof of one such optimization (Free XOR) [KS08] to a State-separating proof (SSP) format [BDF+18], and formally verify the correctness of this converted proof. Formal verification allows us to base the correctness of a protocol or scheme on mathematical and logical constructs, rather than exhaustively testing edge cases, as is often the case in software engineering. We use Domino, a proof verification tool for SSPs, which is currently under development,

and examine the suitability of its current feature set for verifying complex proofs, while also attempting to validate the correctness of our Free XOR State-separating proof.

The initial goal of this project was to verify the Half-Gates garbling scheme [ZRE15]. However, since the Half-Gates scheme works under the Free XOR assumption, it was mandatory for us to first solve the same problem for the Free XOR scheme. Due to the limited timeframe of the project, we were unable to complete both, and leave Half-Gates as future work. We describe the details of this in further sections.

The rest of this paper is organized as follows: Section 2 contains background information on Garbling schemes, optimizations, and Domino. Section 3 details the converted pen-and-paper proof for Free XOR. Section 4 contains further details about the proof tool Domino, and an implementation of the proof, as well as comments on the tool. Section 5 contains concluding remarks and future work.

Chapter 2

Background

The following sections can be split into their own chapters (if needed).

2.1 Garbled Circuit

A Garbled Circuit can be informally defined as a two party protocol, in which both parties want to evaluate a shared function C together, without revealing their inputs to each other. The two parties participating in the protocol are called the **Garbler**, and the **Evaluator**. The function C needs to be expressed as a boolean circuit if a **Garbled Circuit** protocol is used. Input hiding is achieved using something called **garbling**, which in practice is some form of encryption. Similarly, the final output is initially garbled, and needs to be **ungarbled**, which is then implemented using the corresponding decryption function.

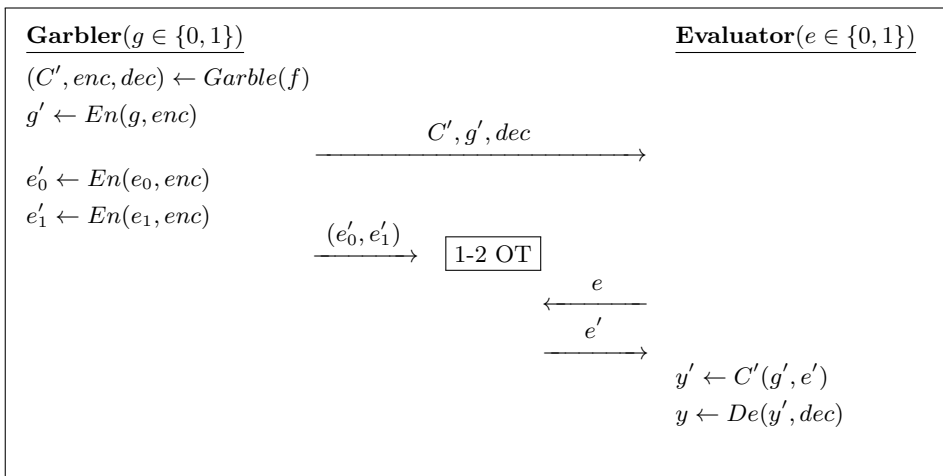


Figure 2.1: Garbled Circuit Protocol

6 2. BACKGROUND

A simple run of the protocol starts with both parties possessing their own input bit respectively (g , and e). The protocol proceeds as follows (ref. Figure 2.1):

1. The Garbler garbles the circuit C to obtain the garbled circuit C' , along with some encoding information (enc) and decoding information (dec). These will be used to garbled and ungarble inputs and outputs.
2. The Garbler garbles its own input g into g' .
3. The Garbler then sends across C' , g and dec to the Evaluator.
4. The Evaluator still needs to garble its own input, but does not possess the encoding information, enc to do so. Therefore, both parties participate in what is called a 1-out-of-2 Oblivious Transfer (OT) protocol [EGL82]. This protocol (adapted for GCs) requires the Garbler to garbled both possible bit values that the Evaluator may possess, and provide them to the OT protocol. Then, the Evaluator provides their chosen bit e to the OT protocol and the OT protocol returns the garbling of that bit (e'). The Garbler remains unaware of which garbling was sent to the Evaluator, and the Evaluator remains unaware of the value of the other garbling.
5. At this point, since the Evaluator possesses both garbled inputs (g' and e') and the Garbled circuit (C'), it can compute the Garbled output, y' by running the garbled inputs through the Garbled circuit.
6. Finally, using the decoding information dec , the Evaluator can obtain y by ungarbling y' . This output is equivalent to the output of the function $C(g, e)$.
7. If necessary, the Evaluator can send the final output back to the Garbler.

The protocol can also be scaled up to support larger inputs from both parties. However, this would increase the complexity of the Oblivious Transfer protocol.

2.1.1 Adversary Model

Both the Garbler and Evaluator are assumed to be *honest-but-curious*. This means that both parties adhere strictly to the protocol specification. However, they try to learn more than what is intended by using the information received from running the protocol, and from their own randomness and state. So, as an example, the Evaluator could try to obtain the Garbler's bit by ungarbling it, but will not succeed since the algorithm dec that the Evaluator receives does not ungarble the inputs.

This kind of adversary is equivalent to the 'Semi-Honest' Adversary described by Aumann and Lindell in [AL10].

2.2 Garbling Schemes

In the previous section, we discussed a 2PC Garbled circuit protocol. However, it is easier to reason about a part of the security of such a protocol using a more generalized construct called a Garbling Scheme. A Garbling Scheme leaves out the Oblivious Transfer sub-protocol and focuses on the security and correctness properties of the garbling of the circuit and input.

A Garbling scheme [BHR12] can be expressed as a 5-tuple of algorithms

$$(\text{Garble}(\text{Gb}), \text{Encode}(\text{En}), \text{Decode}(\text{De}), \text{GarbledEval}(\text{Ev}), \text{Eval}(\text{ev}))$$

1. $\text{Garble}(\text{Gb})$ takes as input the function f and the security parameter 1^λ and returns (F, e, d) , where F is the garbled circuit, e is the encoding information and d is the decoding information.
2. $\text{Encode}(\text{En})$ takes as input the initial input x and the encoding information e , and returns the garbled input X .
3. $\text{GarbledEval}(\text{Ev})$ takes as input the garbled circuit F and the garbled input X , and returns a garbled output Y .
4. $\text{Decode}(\text{De})$ takes input the garbled output Y and the decoding information d , and returns the ungarbled output y .
5. $\text{Eval}(\text{ev})$ takes as input the function f and the plain input x and returns $f(x)$. We will just substitute this with $f(x)$.

Relating this to the representation of the Garbled Circuit protocol in the previous section, the inputs g and e from the previous section are both shares of x . That is, x can be expressed as $x = g \parallel e$.

2.2.1 Properties

The correctness property of a garbling scheme describes its intended behavior. It says that if $(F, e, d) \leftarrow \$ \text{Gb}(1^\lambda, f)$,

$$\forall x, \text{De}(\text{Ev}(F, \text{En}(x, e)), d) = f(x)$$

Besides correctness, a Garbling scheme can have one or more of the following security properties [BHR12]:

1. **Privacy:** Knowledge of (F, X, d) does not reveal anything beyond $f(x)$. This means that it is impossible for an Evaluator to obtain the initial input of the Garbler.

2. **Obliviousness:** Knowledge of (F, X) does not reveal anything. This means that despite being able to solve the garbled circuit, the solver would not be able to map the garbled output to the real output.
3. **Authenticity:** Knowledge of (F, X) does not allow one to find a $Y' \neq Ev(F, X)$ such that a d exists to solve $(De(d, Y') \neq \perp)$.

Of these properties, **Privacy** is the one that is most important to us in this work, and it is what we will continue working with in the rest of the paper.

2.2.2 Defining Privacy

In [BHR12], Bellare, Hoang and Rogaway define privacy as two different cryptographic games: an *indistinguishability* based one (Game $\text{PrvInd}_{G,\Phi}$), and a *simulation* based one (Game $\text{PrvSim}_{G,\Phi,S}$) as described in Figure 2.2.2. In these notations, and elsewhere, Φ refers to the side information function, which describes how much information about the circuit is public. In common usage, it is often either the whole circuit (denoted as $\Phi_{\text{circ}}(C)$), the topology of the circuit ($\Phi_{\text{topo}}(C)$), or the size of the circuit ($\Phi_{\text{size}}(C)$).

$\frac{G_{\text{PrvInd}_{G,\Phi}}^b}{\text{GARBLE}(f_0, f_1, x_0, x_1)}$	$\frac{G_{\text{PrvSim}_{G,\Phi,S}}^b}{\text{GARBLE}(f, x)}$
if $\Phi(f_0) \neq \Phi(f_1)$ then return $\perp$ if $f_0(x_0) \neq f_1(x_1)$ then return $\perp$ $(F, e, d) \leftarrow Gb(1^\lambda, f_b)$ $X \leftarrow En(x_b, e)$ return (F, X, d)	if $b = 0$ then $(F, e, d) \leftarrow Gb(1^\lambda, f)$ $X \leftarrow En(x, e)$ else $y \leftarrow ev(f, x)$ $(F, X, d) \leftarrow \mathcal{S}(1^\lambda, y, \Phi(f))$ return (F, X, d)

Figure 2.2: PrvInd and PrvSim Games - adapted from [BHR12]

In the indistinguishability based game (Game $\text{PrvInd}_{G,\Phi}^b$), the **GARBLE** oracle takes as parameters two pairs of functions and corresponding inputs, (f_0, x_0) and (f_1, x_1) such that $f_0(x_0) = f_1(x_1)$, and the bit b that defines which (f, x) pair **GARBLE** will use. A distinguishing adversary $\mathcal{D}$ tries to guess the bit value b that was used to generate the (F, e, d) tuple. A **INITIALIZE** procedure samples the value of b and it is unknown to the adversary. **GARBLE** then uses the selected (f, x) to generate F , X and d and returns it to the adversary. The advantage of the distinguishing adversary here is defined as the difference in the probability that the adversary

guesses the correct bit and the probability of guessing randomly ($P = 1/2$).
If b' is $\mathcal{D}$'s guess,

$$\text{Adv}_{D, G_{\text{PrivInd}^0}, G_{\text{PrivInd}^1}}(\lambda) = |\Pr[b' = b] - \frac{1}{2}|$$

rewrite next 2 paras for clarity

However, most follow-up works such as Half-Gates [ZRE15] and Free XOR [KS08; App16], use the simulation based paradigm (the PrivSim game from Figure 2.2.2). Since privacy here means that the running the garbling and encoding algorithms Gb and En does not reveal anything about f or x , we let the simulator $\mathcal{S}$ access everything that someone participating in the protocol (semi-honest party) would be able to see, i.e., the security parameter (1^λ), the output of the algorithm y , and the information function of the circuit $\Phi(f)$. Then, this simulator tries to generate a $(\bar{F}, \bar{X}, \bar{d})$ tuple whose distribution is computationally indistinguishable from the one generated by actually running Gb and En .

If the distribution of $(\bar{F}, \bar{X}, \bar{d})$ is actually indistinguishable to an adversary $\mathcal{A}$ from (F, X, d) , an adversary cannot claim to have recovered anything more than what the simulator knew, and hence privacy of f (except $\Phi(f)$) and x is preserved. For further reading on simulators, Lindell provides a comprehensive overview of simulators for beginners in [Lin17].

2.3 Optimizations to Yao's Garbling scheme

2.3.1 Point and Permute

How many bits are supposed to be used for the wire "color"? If (x, y) are all the inputs to the circuit (if there are more than 2 inputs, x is $x_1x_2x_3\dots$ and similarly for y , i.e., for each binary input label x and y , $\text{len}(x) = 2^n$ where n is the number of x inputs. similarly for y). So X and Y are the inputs domain for the two parties, and there need to be $\lceil \log(X) \rceil$ color bits after the key to preserve key length to be equal to the security parameter, while still having the ability to point to a specific X bit on the garbled truth table [EKR18].

2.3.2 Garbled Row Reduction - 3

Compatible with Free XOR

2.3.3 Free XOR

2.3.4 Garbled Row Reduction - 2

2.3.5 FleXOR

2.3.6 Half-Gates

Mention that this was the original objective of the thesis, however work with this requires freexor to be completed

2.3.7 Garbled Gadgets

2.4 Formal Verification and Tools

Start from Formal verification in general (read more too), and then go into tools used for formal verification in cryptography.

2.4.1 Easycrypt

2.4.2 Domino

How is it different from easycrypt? why did we choose this? what is a state separating proof and why is this more suited to those?

Chapter 3

Pen-and-Paper State-Separating Proof

Read half gates and chris's paper to understand what to do here even. (WIP)

3.1 State Separating Proofs

- What are games and packages
- Something about why state separating proofs exist. "composable" and "constructive" security
- State separating proofs allow us to make intermediate arguments (hybrids) in a reusable manner. It is often more intuitive to model something as its own package that has inputs and outputs than to have it be part of many different codes and realise that its the same thing.

Composability of layers is very important so we can derive circuit security from just proving layer security. The simulator for layer n must output a garbling of its output which is then supplied to a new simulator for layer $n+1$. Therefore, if a simulator can simulate a gate by getting its inputs (garbled input) and return garbled output, it can be composed with the simulator for the next gate and so on. Same idea for layers.

Actually, to add to the above, the simulator does NOT know the input. It only gets the circuit information loss of the garbled circuit $\Phi(C)$, the output of the garbled circuit Y , and the security parameter 1^λ . So it samples a random input label (in this case its the label, in the original Yao paper it was a random bitstring sampled for the wire key, but since we are working in a setting where the Free XOR optimization is assumed, we will probably end up sampling a random Δ and sharing that state across simulator invocations to generate valid labels for the wires.)

3.2 Existing Half-Gates Proof and how it is to be modelled into state separating proofs

Chapter 4

Domino

4.1 What is Domino?

What is domino and how does it work? The layers that exist, and the design in general.

Domino is a proof assistant for State-separating proofs (SSPs) that allows the user to validate a SSP using familiar synatax, and an underlying SMT solver. It uses very similar syntax and constructs as SSPs, such as packages, games and theorems. The explicit relation between these games is defined using invariant files, written in the SMT-LIB language. These games are then checked against different lemmata that are needed to fulfil the relation that the user wants to establish.

Domino is published as a rust package available on Github. The domino program/crate itself contains a compiler that has its own defined language that intentionally tries to be very similar the SSP syntax defined in [BDF+18].

4.2 Using Domino

4.2.1 Dependencies

Since Domino is published as a Rust package, it requires a Rust toolchain, which can be installed using *rustup*. Rustup manages multiple Rust toolchains, and is the most straightforward way to getting set up on Linux. Rustup is typically installed using a bash script from the official website, but it is also available on most system package managers. It can also be built from source. Installing rustup automatically installs the Rust binary and cargo (the package manager), among other components. These can also be obtained without rustup, and the installation page explains the methods in more detail [Rus].

The second prerequisite to run Domino is a functioning CVC5 installation, and for the *cvc5* executable to be added to PATH, so that Domino can invoke it while

running. CVC5 can be built from source following the instructions in [CVC]. CVC5 is the SMT solver that checks whether the constraints given by our proof are satisfiable or not. This is done in practice by taking the negation of the condition we want, and checking for satisfiability of that negation. If it exists, the condition is false.

4.2.2 Project Structure

A Domino project directory requires the presence of an empty *ssp.toml* file. It allows SSP logic to be written into packages, games and theorem files. These are all placed under their own directories, i.e., folders called *packages*, *games*, and *theorem*.

Each of these folders can then contain one or more files. Domino checks packages and games for syntactic correctness even if they are not used in a theorem. This is intentional design, since when writing state separating proofs, the components (packages and games) are usually set up before the actual proof.

In older versions of Domino, theorems were called proofs, and the folder and file header for proofs were named as such.

An Example Theorem in Domino

mIND-CPA game [here](#), talk about its packages, and the two games with diagrams, and how the theorem to prove states that the two games are indistinguishable.

4.3 Coding Half-gates in Domino

4.4 Improvements

Chapter 5

Conclusion

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